Digital PAM Channel & Matched Filter

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Abstract:

In digital communication systems, recovering transmitted signals from noisy environments is a fundamental challenge. As part of this project, matched filters are designed and implemented to improve signal detection accuracy when Additive White Gaussian Noise (AWGN) is present. Through the use of both MATLAB code and Simulink, we analyze the performance of 2-level and M-level Pulse Amplitude Modulation (PAM) schemes. Simulations are conducted to measure Bit Error Rate (BER) and Symbol Error Rate (SER) across a range of Signal-to-Noise Ratio (SNR) values. Matching filters significantly improve detection performance and decreasing symbol spacing increases BER and SER by increasing modulation levels (M). Both theoretical principles and practical simulations are addressed to provide a comprehensive understanding of correlation receivers and their effectiveness in digital communication channels.

Introduction:

Digital communication has become an essential aspect of modern information systems, driving the need for faster and more reliable data transmission. A core challenge in this domain is the recovery of transmitted signals that have been distorted by noise, particularly in the form of Additive White Gaussian Noise (AWGN). Correlation receivers and matched filters serve as optimal tools for detecting known waveforms and maximizing Signal-to-Noise Ratios (SNRs)

Matched filtering is mathematically proven to be optimal for detecting a known signal embedded in noise. Using time-reversed version of the expected waveform, it effectively extracts the signal component while minimizing noise interference. This technique is especially critical in digital modulation schemes such as Pulse Amplitude Modulation (PAM), where binary or M-ary digital symbols are represented by varying amplitude levels.

This project simulates and evaluates the performance of matched filters in detecting PAM signals under noisy conditions. In both Simulink and MATLAB, we plot the BER and SER as a function of SNR for a variety of PAM levels (2, 4, 8, etc.). Through these simulations, we can see how matched filters can mitigate the effects of noise on communication systems. As a result, this report examines matched filters in digital communication in both a theoretical and practical perspective, using quantitative results to assess their impact on errors.

Theoretical Background:

- **Matched Filtering and Correlation Receivers:**

In digital communication, the received signal is often corrupted by noise, making it difficult to detect the original message. A **correlation receiver** is a device that compares the received signal with a known signal (called a *template*) to check how closely they match. It does this by multiplying and adding the signals over time. An SNR filter maximizes Signal-to-Noise Ratio (SNR) at the receiver. It uses the known waveform as a filter by flipping it in time. As a result, the incoming signal is processed with this time-reversed signal. Even in the presence of noise, both methods help detect the signal accurately.

Key idea: Matched filtering is the best method to detect a known signal in white noise.

➢ BER, SER, and SNR:

- **Bit Error Rate (BER)**: This is the percentage of bits that are received incorrectly. For example, if 1000 bits are sent and 10 are received wrong, the BER is 1%.
- **Symbol Error Rate (SER)**: Instead of individual bits, we sometimes look at symbol errors. A symbol may represent more than one bit (like in 4-PAM or 8-PAM), and SER tells us how often a whole symbol is received incorrectly.
- **Signal-to-Noise Ratio (SNR)**: This tells us how strong the signal is compared to the noise. A higher SNR means better quality and fewer errors. It is usually measured in decibels (dB).

➢ System Design:

A digital communication system is made of several blocks that work together to send and receive data. Here's a breakdown of the system used in this project:

- **Pulse Amplitude Modulation (PAM):**
  is a method of sending digital data by changing the amplitude (height) of each pulse.
  - **2-PAM** uses two levels (e.g., -1 and +1) to represent binary 0 and 1.
  - **M-PAM** (e.g., 4-PAM, 8-PAM) uses more levels to send more bits per symbol.

For example:

- ❖ 2-PAM sends 1 bit per symbol
- ❖ 4-PAM sends 2 bits per symbol
- ❖ 8-PAM sends 3 bits per symbol

More levels (higher M) mean faster data rates, but the symbols are closer together and easier to confuse with noise.

- **Block Diagram of the System:**

  The system used in the simulation includes the following blocks:

  1. **Transmitter:**
     - Generates random digital bits
     - Maps bits to PAM symbols
     - Sends the symbols over the channel
  2. Channel (AWGN)
     - Adds random noise (Gaussian noise) to the signal
     - The noise simulates real-world interference
  3. Matched Filter (Receiver)
     - Processes the received signal
     - Matches it to the expected shape of the original signal
     - Improves the signal and reduces the noise

4. Decision Device:

- Decides which symbol was most likely sent
- Converts the received signal back to bits
- Compares received bits to original bits to calculate errors

➢ MATLAB Simulations:

The purpose of this section is to evaluate the performance of matched filters with various pulse amplitude modulation (PAM) levels using MATLAB simulations. BER and the modulation level (M) of system performance were studied in relation to Signal-to-Noise Ratio (SNR).

- BER vs. SNR for M-PAM:

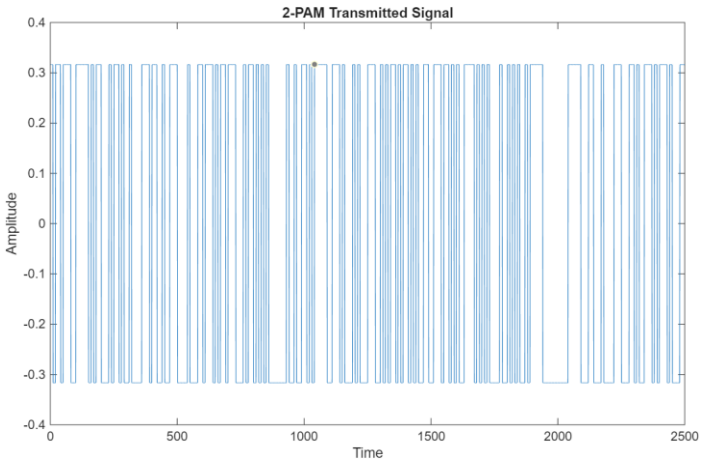


Fig1: 2-PAM Transmitted signal

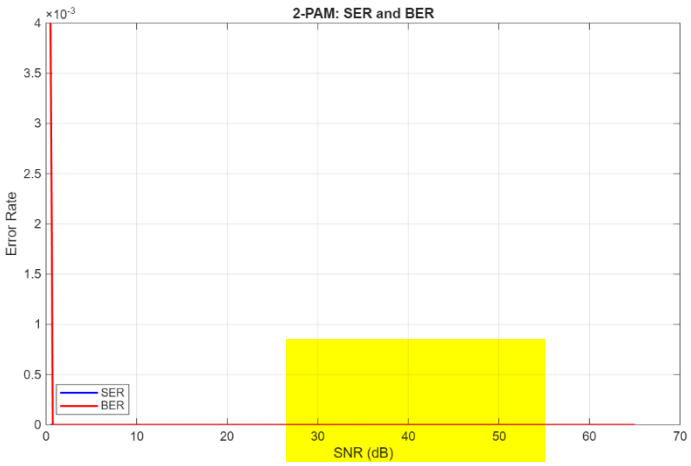


Fig2: The SER and BER Vs the SBR for the 2-PAM signal

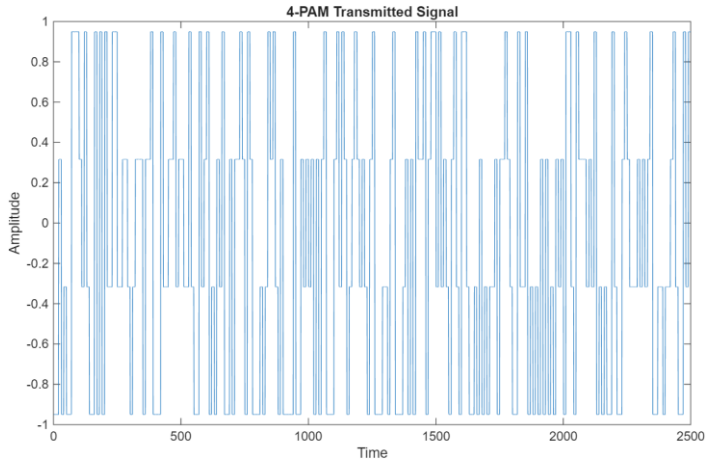


Fig3: 4-PAM Transmitted signal

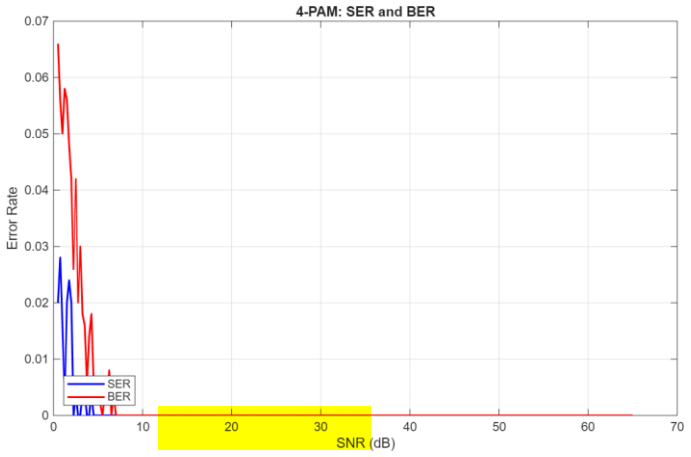


Fig4: The SER and BER Vs the SBR for the 4-PAM signal

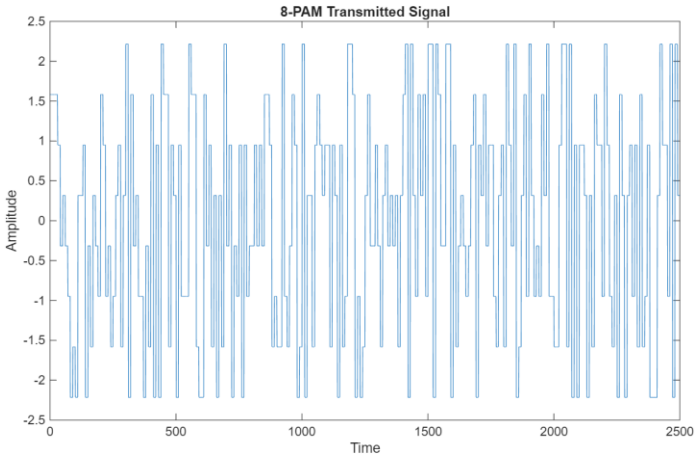


Fig5: 8-PAM Transmitted signal

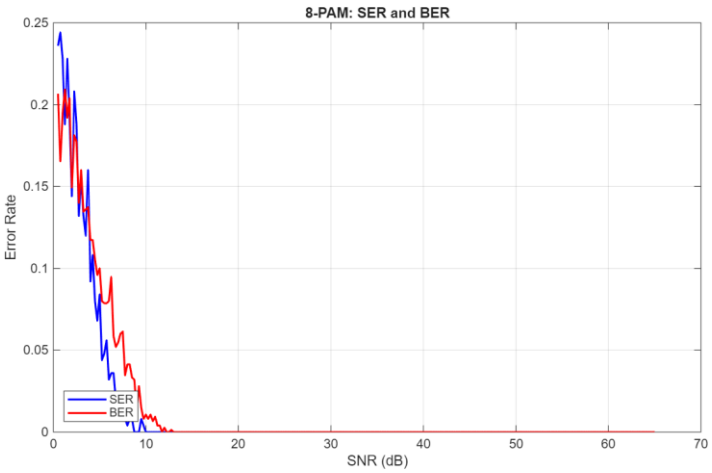


Fig6: The SER and BER Vs the SBR for the 8-PAM signal

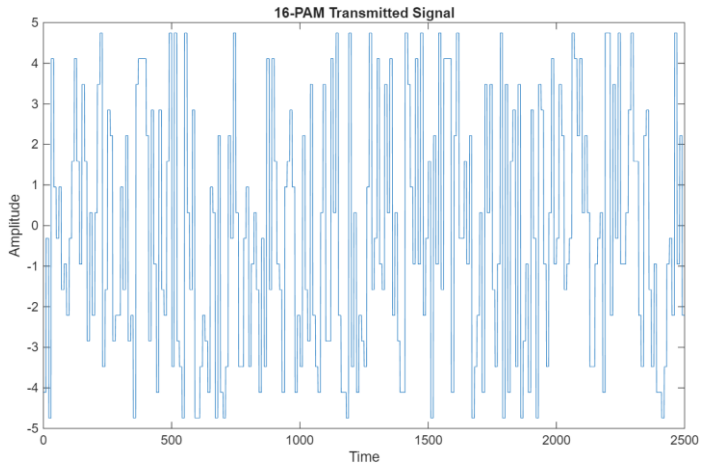


Fig7: 16-PAM Transmitted signal

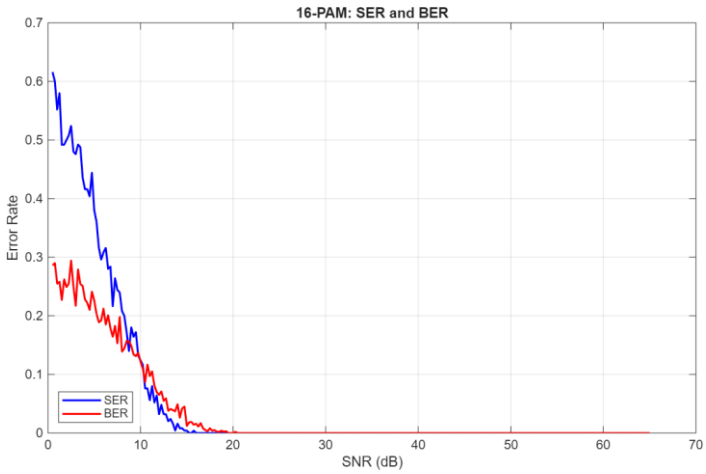


Fig8: The SER and BER Vs the SBR for the 16-PAM signal

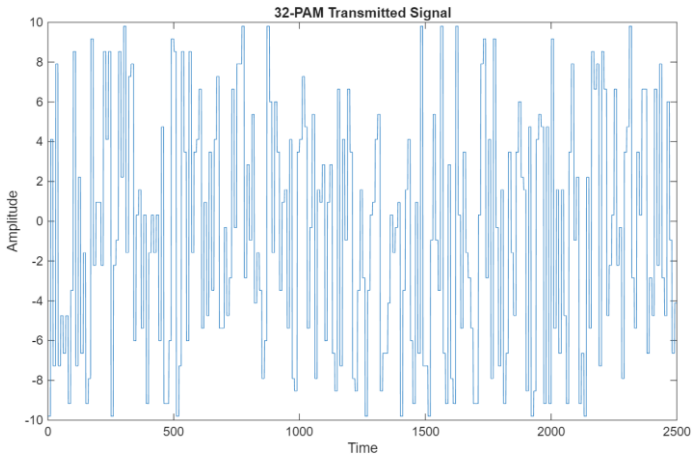


Fig9: 32-PAM Transmitted signal

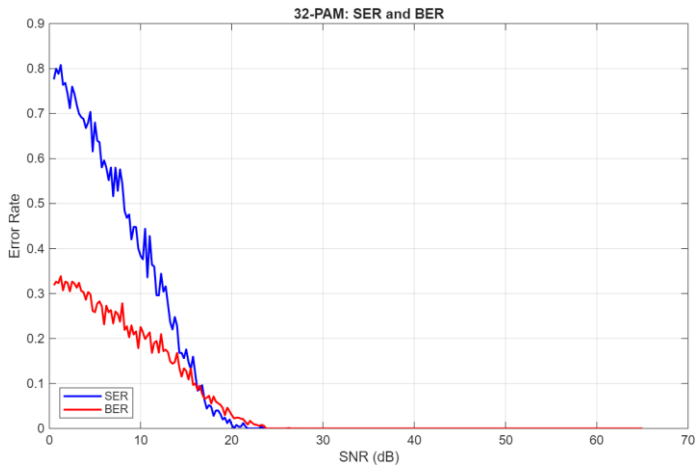


Fig10: The SER and BER Vs the SBR for the 32-PAM signal

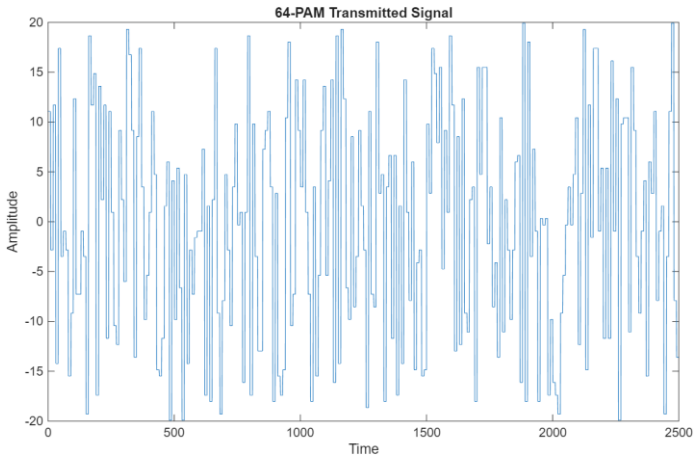


Fig11: 64-PAM Transmitted signal

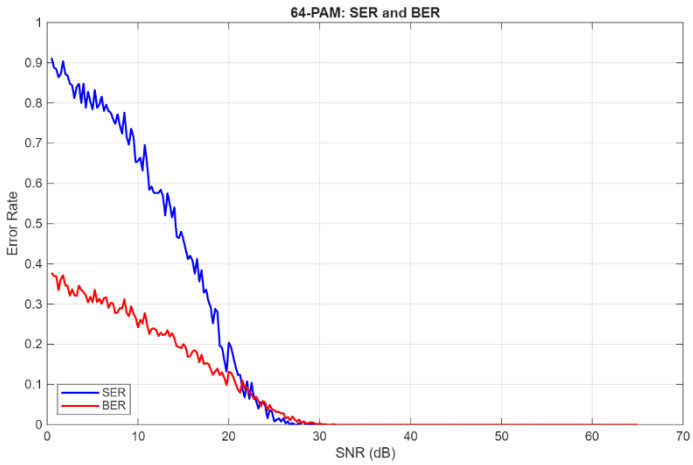


Fig12: The SER and BER Vs the SBR for the 64-PAM signal

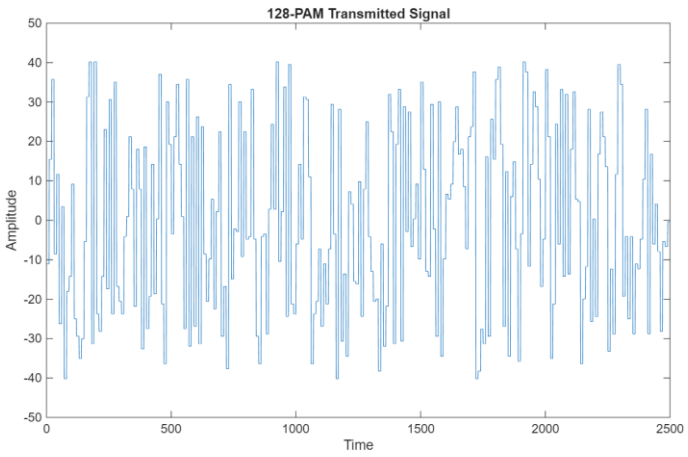


Fig13: 128-PAM Transmitted signal

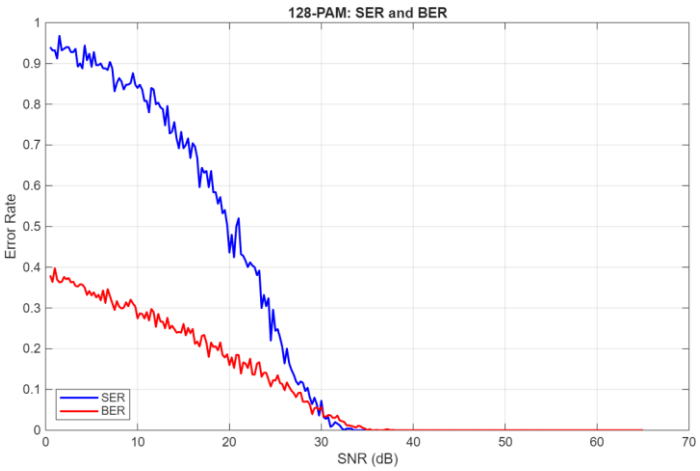


Fig14: The SER and BER Vs the SBR for the 128-PAM signal

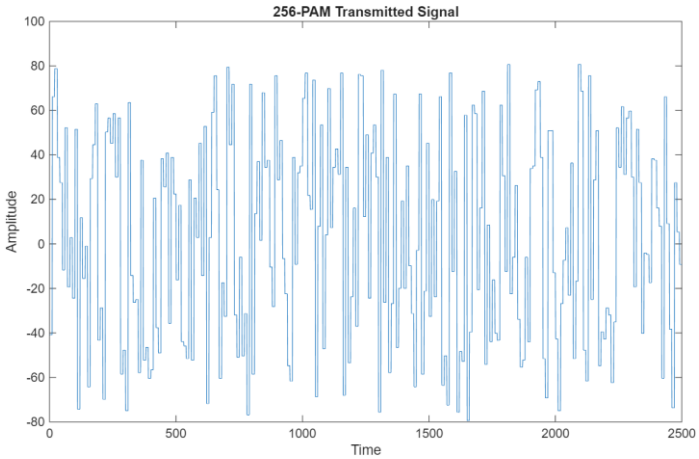


Fig15: 256-PAM Transmitted signal

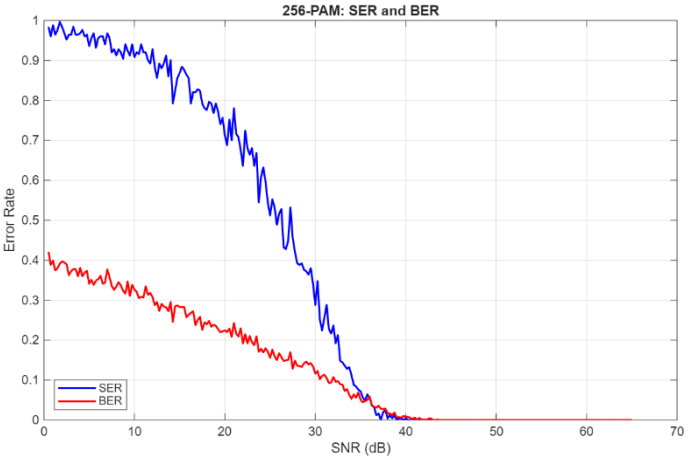


Fig16: The SER and BER Vs the SBR for the 256-PAM signal

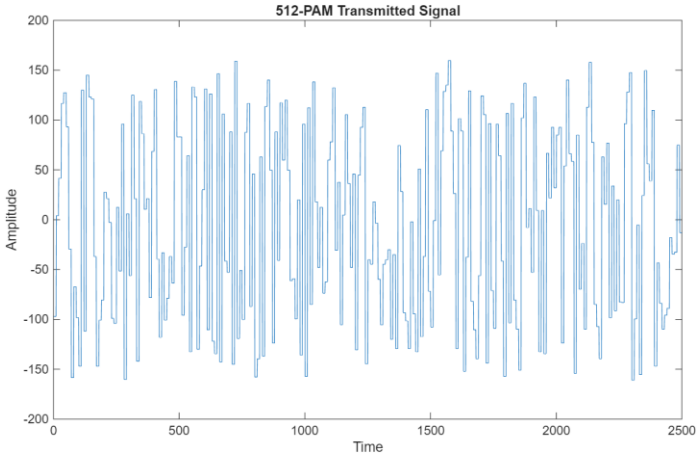


Fig17: 512-PAM Transmitted signal

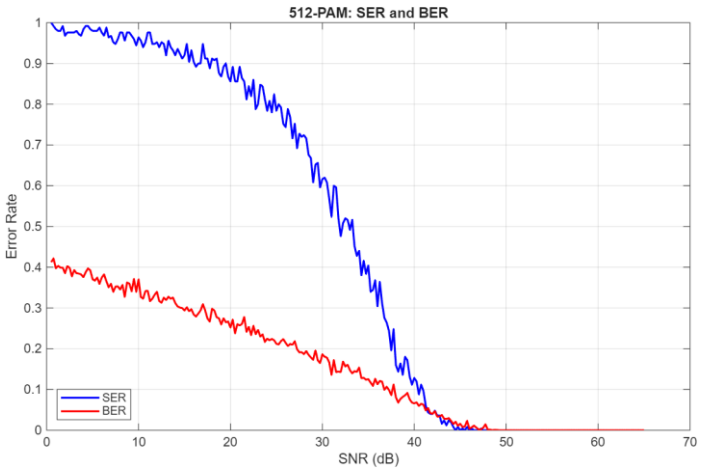


Fig18: The SER and BER Vs the SBR for the 512-PAM signal

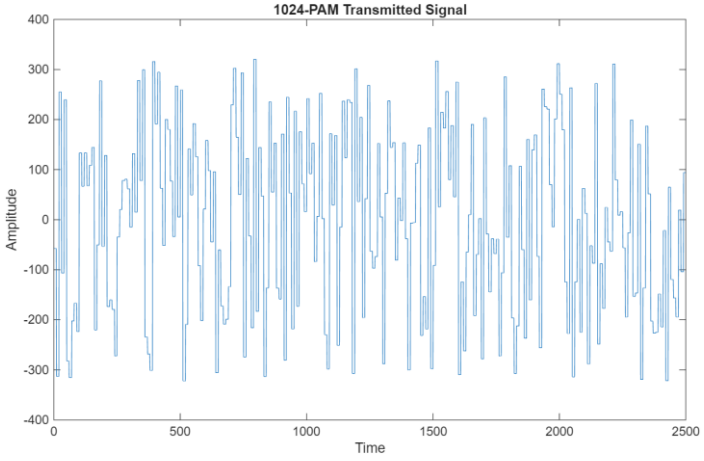


Fig19: 1024-PAM Transmitted signal

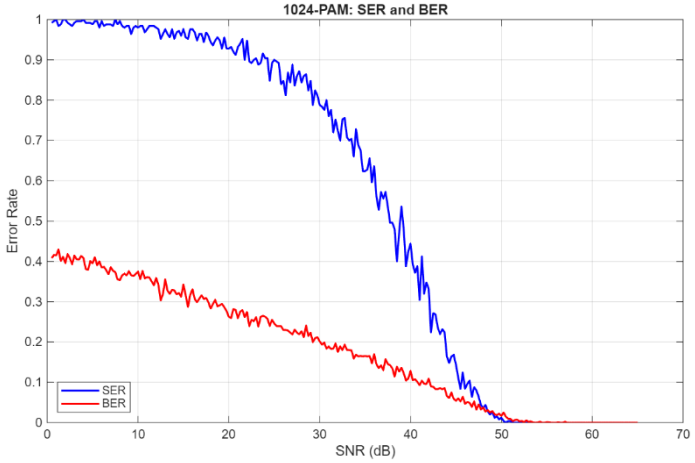


Fig20: The SER and BER Vs the SBR for the 1024-PAM signal

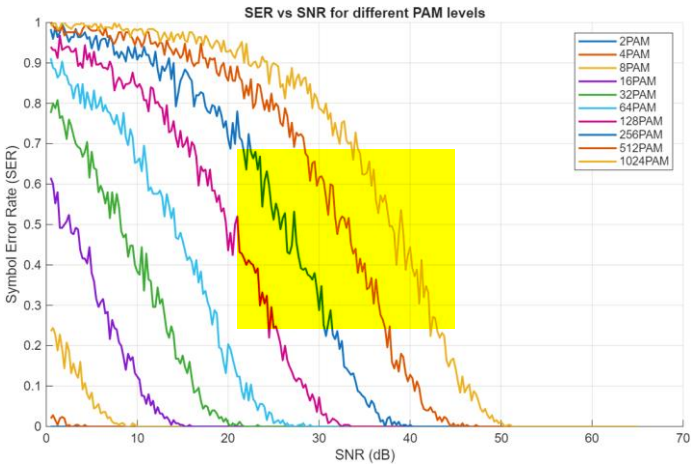


Fig21:SER Vs SNR for different PAM levels

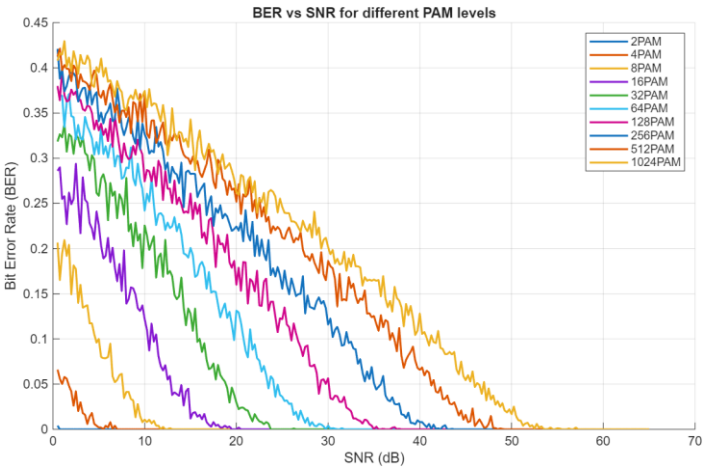


Fig22:BER Vs SNR for different PAM levels

➢ Simulink Modeling:

Our objective in this section was to visually simulate the digital communication system and compare the performance of 2-PAM and 4-PAM modulation schemes in a noisy environment using MATLAB Simulink.

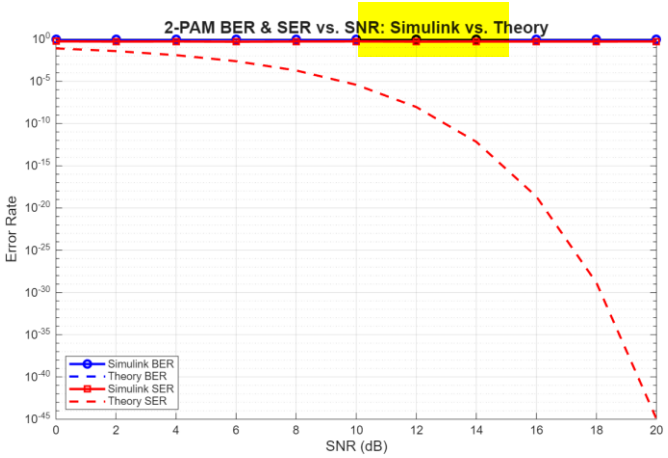


Fig23:2-PAM BER& SER Vs. SNR; Simulink Vs. Theory

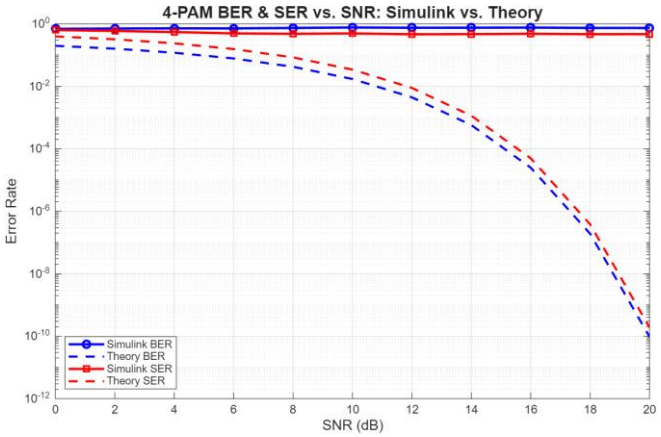


Fig24:4-PAM BER& SER Vs. SNR; Simulink Vs. Theory

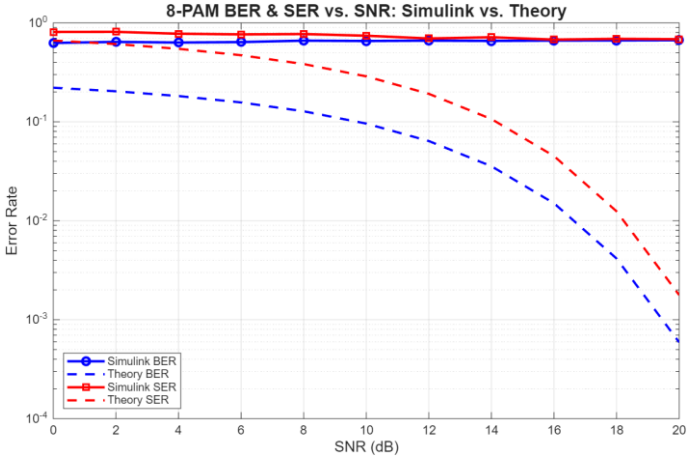


Fig24:8-PAM BER& SER Vs. SNR; Simulink Vs. Theory

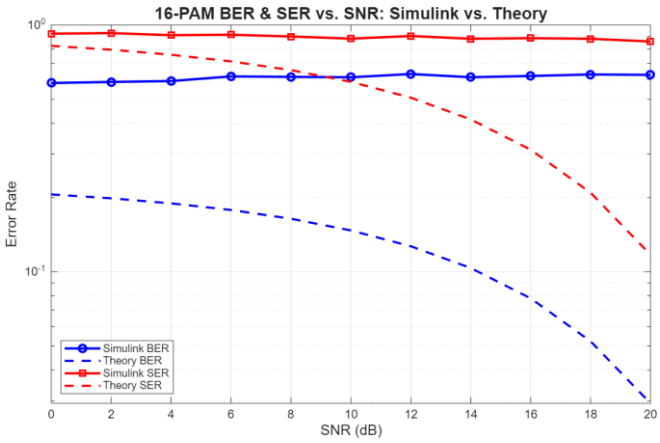


Fig25:16-PAM BER& SER Vs. SNR; Simulink Vs. Theory

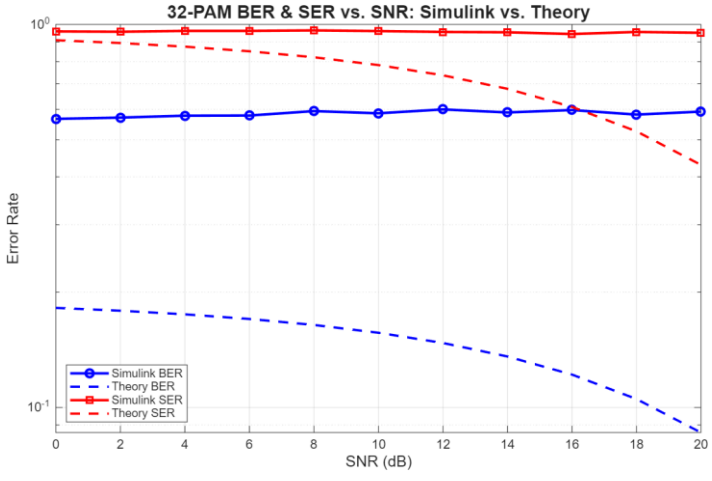


Fig26:32-PAM BER& SER Vs. SNR; Simulink Vs. Theory

➢ Analysis and Discussion:

As a result of running both the MATLAB simulations and Simulink models, we analyzed the effect of different modulation levels and noise conditions on the performance of our system.

- Effect of Modulation Level (M):
  From both BER and SER results, we observed that:
  - Lower modulation levels (like 2-PAM) had lower error rates. As a result, it is easier to distinguish the correct symbol since the difference between signal levels is greater.
  - Higher modulation levels (like 16-PAM) Due to the closer line spacing and greater sensitivity to noise, error rates were higher.
- Effect of SNR:
  - As **SNR increases**, the **BER and SER decrease** for all M-PAM levels. A higher SNR translates into stronger signal compared to noise, so the original message can be recovered more easily.
  - At high SNR values (above 6 dB), Under clean conditions, 2-PAM exhibits excellent performance with nearly zero BER.
- MATLAB vs Simulink:
  - We were able to test a range of M and SNR values with the MATLAB code because it gave us precise numerical results.
  - Using Simulink, we visualized how noise affects signal quality using eye diagrams and confirmed that 2-PAM signals are more robust than 4-PAM signals.
- Matched Filter Impact:
  In all simulations, the **matched filter** played a crucial role in improving detection. As a result, the received signal was aligned with the expected pulse shape, which reduced the effect of noise. A matched filter can improve the performance of a communication system by increasing the signal-to-noise ratio.

References:

[1] Apena, W. O. (n.d.). Performance evaluation of PAM-4 and pam-2 modulation techniques using MATLAB. European Journal of Engineering and Technology Research. https://www.ejers.org/index.php/ejers/article/view/679.

[2] Matched filters. Matched Filters- an overview | ScienceDirect Topics. (n.d.). https://www.sciencedirect.com/topics/physics-and-astronomy/matched-filters.

[3] "BER Performance Comparison of M-PAM over AWGN and Fading Channels," 2021 IEEE 1st International Maghreb Meeting of the Conference on Sciences and Techniques of Au tomatic Control and Computer Engineering MI-STA, 2021, pp. 802-806, doi: 10.1109/MI STA52233.2021.9464451. Retrieved from: https://ieeexplore.ieee.org/document/9464451.